Submitted by

Chapter 1

Introduction

1.1 Background and Motivation

Public sector institutions in Thailand operate under a strict and multi-layered framework of financial regulations. At King Mongkut's Institute of Technology Ladkrabang (KMITL), administrative and finance personnel are required to follow the Government Procurement and Supply Management Act B.E. 2560, the Ministry of Finance Procurement Regulations, the Fiscal Discipline Act B.E. 2561, and a collection of institutional operational manuals. These documents collectively govern every aspect of how the university plans, approves, executes, and records its budget expenditures.

Despite the availability of these documents, accessing the right information at the right moment remains a persistent challenge. The regulatory corpus is large, spanning over a dozen major PDF files written primarily in Thai, and the content is both technical and deeply procedural. A staff member who needs to verify a procurement threshold, confirm a required attachment for an expense claim, or understand the correct e-Market process for a given purchase value must manually search through multiple documents, often without a clear starting point. This process is not only time-consuming but also prone to human error, since regulations are frequently referenced across different files and procedural steps can be easily overlooked.

In recent years, the emergence of large language models (LLMs) and Retrieval-Augmented Generation (RAG) systems has opened a practical path toward automating this kind of document-based question answering. Unlike a simple keyword search, a RAG system understands the meaning of a question, retrieves the most relevant passages from a document corpus, and generates a coherent, cited answer in natural language. This makes it well-suited to regulatory environments where questions are often nuanced and answers must be traceable to official sources.

Motivated by these developments, this project proposes the development of a Thai-language AI chatbot specifically designed for KMITL's budget disbursement and procurement

domain. The system is intended to serve as an always-available, accurate, and source-cited assistant for institutional staff, reducing their reliance on manual document searches and helping them carry out financial procedures with greater confidence and correctness.

1.2 Problem Statement

The central problem this project addresses can be summarised in three interconnected dimensions, each of which directly affects the operational efficiency and regulatory compliance of KMITL's administrative functions.

First, budget disbursements are frequently delayed. When staff encounter an unfamiliar procedure or an ambiguous regulation, they must pause their work to search manually through the relevant documents or consult a senior colleague. This search process is rarely straightforward, as the answer to a single question may require cross-referencing information from multiple files. The resulting delays accumulate across teams and departments, slowing down procurement cycles and disrupting project timelines.

Second, disbursements are sometimes carried out incorrectly. The complexity of the regulatory framework means that even experienced personnel occasionally misinterpret a rule, use an outdated procedure, or omit a required supporting document. These errors can lead to rejected reimbursement claims, requests for revision, and in more serious cases, compliance violations that carry institutional consequences. The root cause is not a lack of diligence but a lack of convenient access to reliable, up-to-date procedural guidance.

Third, the information itself is difficult to locate. KMITL's regulatory and procedural knowledge is stored exclusively in PDF files, with no unified search interface, no structured indexing, and no mechanism for staff to ask questions in plain language. Newer employees, in particular, have no efficient way to build familiarity with the documents and must rely heavily on institutional knowledge passed down informally, which is neither scalable nor consistently accurate.

Together, these three problems point to a clear need for a system that makes regulatory knowledge immediately accessible, contextually accurate, and verifiably grounded in official documents.

1.3 Objectives

This project is guided by the following specific objectives:

1. To design and implement a Retrieval-Augmented Generation (RAG) pipeline capable of ingesting, indexing, and retrieving information from KMITL's Thai-language regulatory PDF documents.
2. To develop a conversational chatbot interface that allows institutional staff to pose questions in natural Thai and receive accurate, clearly reasoned answers in return.
3. To ensure that every system response includes explicit citations identifying the source document and page number, enabling users to verify answers against the original regulatory text.
4. To evaluate the system's retrieval quality using standard information retrieval metrics, including Recall, Precision, and Mean Reciprocal Rank, and to assess end-to-end answer quality using an LLM-as-a-Judge framework modelled on the RAGAS methodology.
5. To deliver a functional and deployable solution that operates at zero ongoing cost by leveraging locally-hosted embedding models and a free-tier language model API.

1.4 Scope of the Project

This project is focused specifically on the budget disbursement and procurement domain of KMITL. The knowledge base is constructed from a defined set of official regulatory and operational documents, as described in Chapter 4. The system is designed for use by KMITL administrative and finance personnel and is not intended as a general-purpose legal or financial advisory tool.

The following boundaries define the scope of this work:

Dimension	Scope
Domain	Budget disbursement and procurement procedures at KMITL
Language	Thai-language input and output; regulatory documents are primarily in Thai
Document Sources	Defined set of official PDFs, covering laws, ministerial regulations, and KMITL operational manuals
Users	KMITL administrative and finance personnel

Dimension	Scope
Deployment	Streamlit web application, intended for internal institutional use
Not in Scope	Real-time data integration, financial transaction processing, legal advice, or documents outside the defined corpus

The system does not connect to any live financial system or database. It provides informational guidance only, and users are expected to refer to official documents and consult responsible officers for final decisions on any consequential financial matter.

1.5 Expected Outcomes

Upon successful completion of this project, the following outcomes are anticipated:

- A fully functional RAG-based chatbot application, deployable via Streamlit, that provides Thai-language answers to questions about KMITL procurement and disbursement procedures.
- A vector database built from KMITL's regulatory PDF corpus, enabling fast and semantically accurate document retrieval.
- Measured retrieval performance with a Mean Reciprocal Rank (MRR) above 0.30 and a cosine similarity score above 0.60, confirming that the system consistently locates relevant content.
- End-to-end answer quality scores of at least 4.0 out of 5.0 across faithfulness, answer relevance, and context precision as assessed by the LLM-as-a-Judge evaluation framework.
- A practical reduction in the time required for staff to locate procedural guidance, with all answers accompanied by verifiable source citations.
- A replicable evaluation methodology, including a golden dataset and automated scoring scripts, that can be extended as the document corpus grows.

1.6 Contributions

This project makes the following contributions to both the institution and the broader field of applied natural language processing:

From a practical standpoint, the system addresses a genuine operational gap at KMITL by making institutional regulatory knowledge queryable in natural language. To the best of the authors' knowledge, no comparable AI-assisted document retrieval system currently exists for KMITL's procurement and disbursement domain. The system is also designed to be maintainable and extensible, so new documents can be added to the corpus and the vector database rebuilt without modifying the core application code.

From a technical standpoint, the project demonstrates the feasibility of building a high-quality, zero-cost RAG system for a low-resource language (Thai) in an institutional regulatory context. The use of a locally-hosted multilingual embedding model, instead of a paid embedding API, ensures that the system remains cost-free regardless of query volume. The integration of an LLM-as-a-Judge evaluation framework adapted from the RAGAS methodology provides a principled and reproducible approach to assessing answer quality in the absence of large human-annotated evaluation sets.

Finally, the project contributes a curated golden dataset of representative procurement and disbursement questions, which can serve as a foundation for future evaluation and benchmarking as the system is extended.

Chapter 6

Retrieval Evaluation Report

6.1 Executive Summary

This chapter presents a comprehensive evaluation of the retrieval component of the Retrieval-Augmented Generation (RAG) system deployed for procurement-domain question answering. The system employs the sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 embedding model and retrieves the top K = 6 document chunks per query. Evaluation was conducted on a curated golden dataset of 323 question-answer pairs spanning a wide range of procurement regulation topics.

Across all four retrieval metrics, the system demonstrates strong and consistent performance. Recall@6 of 87.31 percent confirms that the relevant chunk is surfaced within the top six results for the vast majority of queries. A Mean Reciprocal Rank of 0.7406 indicates that, when the relevant chunk is found, it typically appears near the top of the ranked list. These results validate the suitability of the chosen embedding model for multilingual, domain-specific retrieval.

87.31% **Recall @ 6** avg over 323 questions	**62.95%** **Precision @ 6** avg over 323 questions	**0.7406** **MRR** max = 1.0	**0.7113** **Avg Cosine** max = 1.0

6.2 Evaluation Methodology

The evaluation follows a standard information retrieval (IR) framework adapted for RAG systems. For each question in the golden dataset, the retrieval pipeline returns the K = 6 highest-scoring document chunks as ranked by cosine similarity. These retrieved chunks are then compared against the ground-truth relevant chunk(s) associated with each question.

The assessment covers four dimensions:

- Coverage — whether the relevant chunk appears anywhere within the top-K results (Recall@K).
- Signal-to-noise ratio — the proportion of retrieved chunks that are genuinely relevant (Precision@K).
- Ranking quality — how highly the first relevant chunk is ranked (MRR).
- Semantic alignment — the average cosine similarity between query and retrieved chunks (Avg. Cosine Score).

6.3 Golden Dataset Construction

The golden dataset comprises 323 question-answer pairs drawn from the Thai government procurement domain, covering topics such as price determination procedures, e-Market and e-Bidding processes, exemption criteria, and asset management regulations. Each question is paired with a ground-truth expected answer and the corresponding authoritative document chunk from which the answer is derived.

Questions were formulated to reflect realistic user queries submitted to the RAG system in production, including both straightforward definitional questions and multi-step procedural enquiries. This diversity ensures that the evaluation captures both the breadth and the depth of retrieval capability.

6.4 Retrieval Evaluation Results

6.4.1 Metrics Definition and Rationale

The four metrics used in this evaluation are defined as follows:

- Recall@K measures whether at least one relevant chunk appears within the top K retrieved results. A score of 1 indicates the relevant chunk was found; 0 indicates it was absent. High recall is a prerequisite for downstream answer quality, as the language model can only generate correct answers from chunks that are actually retrieved.
- Precision@K measures the proportion of the K retrieved chunks that are relevant to the query. Lower precision implies that the retrieval set contains more noise, which may distract the language model or dilute the signal from the truly relevant passage.
- Mean Reciprocal Rank (MRR) is the average of the reciprocal of the rank position at which the first relevant chunk appears. An MRR of 1.0 means the relevant chunk is always ranked first; values approaching 0 mean it appears late or not at all. MRR is particularly important in RAG systems where context window constraints limit how many chunks can be passed to the generator.
- Average Cosine Similarity measures the mean semantic distance between the query embedding and the retrieved chunk embeddings. Values close to 1.0 indicate high semantic alignment, while lower values may signal vocabulary mismatch or domain adaptation gaps.

6.4.2 Summary of Retrieval Metrics

Table 6.1 presents the overall retrieval metrics aggregated across all 323 evaluation questions.

Metric	Symbol	Result	Threshold / Status
Recall @ K	Recall@6	**87.31%**	≥ 80% ✓

Precision @ K	Precision@6	**62.95%**	≥ 50% ✓
Mean Reciprocal Rank	MRR	**0.7406**	≥ 0.60 ✓
Avg. Cosine Similarity	Cosine	**0.7113**	≥ 0.65 ✓

Table 6.1 — Overall Retrieval Metrics (K = 6, n = 323 questions)

All four metrics exceed the established acceptability thresholds, confirming that the retrieval pipeline is functioning as intended. The embedding model demonstrates strong capacity to surface relevant procurement-domain content in Thai language queries.

6.4.3 Recall@6 Distribution

Recall@6 is a binary metric per question: either the relevant chunk appears in the top six results (Recall = 1) or it does not (Recall = 0). Table 6.2 summarises the distribution across the evaluation set.

Recall Category	No. of Questions	Proportion
Recall = 1.0 (full hit)	282	87.3%
Recall = 0.0 (complete miss)	41	12.7%

Table 6.2 — Recall@6 Distribution (n = 323 questions)

Of the 323 questions, 282 (87.3 percent) achieve perfect recall — meaning the ground-truth chunk is present in the retrieved set. The remaining 41 questions (12.7 percent) represent complete retrieval failures, where the relevant chunk is not surfaced at any rank position within K = 6. These failure cases tend to involve queries with highly specific procedural details, numerical thresholds, or field-level references in electronic procurement forms, where surface-form vocabulary diverges substantially from the source document language.

6.4.4 MRR Distribution

Mean Reciprocal Rank captures not only whether the relevant chunk is retrieved, but also how prominently it is ranked. Table 6.3 breaks down the MRR distribution into four bands.

MRR Band	No. of Questions	Proportion
MRR = 1.0 (top-rank hit)	217	67.2%
MRR 0.50 – 0.99 (early hit)	25	7.7%
MRR 0.01 – 0.49 (late hit)	40	12.4%
MRR = 0.0 (no relevant chunk)	41	12.7%

Table 6.3 — MRR Band Distribution (n = 323 questions)

The dominant outcome is an MRR of 1.0, achieved in 67.2 percent of all questions, meaning the most relevant chunk is returned as the very first result. A further 7.7 percent of questions yield MRR values between 0.50 and 0.99, reflecting cases where the relevant chunk is retrieved within the first two ranks. The 12.4 percent of questions with MRR below 0.50 represent instances where the relevant chunk is found but ranked third or lower, and the 12.7 percent with MRR of zero correspond directly to the complete-miss cases identified in the Recall analysis.

6.4.5 Precision@6 Distribution

Precision@6 measures the density of relevant chunks within the retrieved set. Because each query typically has a single ground-truth chunk, a Precision@6 of 1/6 ≈ 0.167 would be expected if exactly one relevant chunk is returned among six. Higher values may indicate multiple chunks from the same authoritative passage, while lower values may reflect chunk duplication artefacts. Table 6.4 presents the precision distribution.

Precision Band	No. of Questions	Proportion
Precision ≥ 0.80 (high)	152	47.1%
Precision 0.50 – 0.79 (moderate)	84	26.0%
Precision < 0.50 (low)	87	26.9%

Table 6.4 — Precision@6 Band Distribution (n = 323 questions)

The high proportion of questions (47.1 percent) with precision at or above 0.80 is partly attributable to a structural pattern observed in the data: 311 of the 323 questions exhibit duplicate cosine scores among retrieved chunks, indicating that the same or near-identical document chunks are being returned at multiple rank positions. This inflates precision artificially and warrants attention in the chunking and deduplication pipeline. The 26.9 percent of questions with precision below 0.50 reflect genuine noise in the retrieval set.

6.5 Failure Case Analysis

A qualitative examination of the 41 zero-recall questions reveals several recurring failure patterns. Table 6.5 lists a representative sample of queries that produced no relevant chunk within the top six results.

ID	Question (abbreviated)	MRR	Recall	Cosine
Q8	ขั้นตอนการจัดซื้อจัดจ้างวิธีเฉพาะเจาะจง วงเงินเกิน 100,000 บาท	0.00	0	~0.67

Q10	ผู้ถือหุ้นรายใหญ่ตามระเบียบกระทรวงการคลัง พ.ศ. 2560 หมายความว่า	0.00	0	~0.69
Q13	หลักการจัดซื้อจัดจ้างภาครัฐตาม พรบ. 2560 มีอะไรบ้าง?	0.00	0	~0.68
Q30	ข้อกำหนดสำหรับการแนบไฟล์เอกสารราคากลางมีอะไรบ้าง	0.00	0	~0.70
Q34	ในการบันทึกข้อมูลเอกสารส่วน e-market ช่อง E6 E8 E9 คืออะไร	0.00	0	~0.67

Table 6.5 — Representative Zero-Recall Failure Cases

Analysis of these failure cases reveals three primary root causes:

- Vocabulary mismatch: Queries containing formal legal terminology or abbreviated procurement codes (e.g., field identifiers such as E6, E8, E9) do not share sufficient lexical overlap with the source document passages, causing the embedding model to assign low cosine similarity scores to the correct chunk.
- Chunking granularity: Several high-level principle-based questions (e.g., Q13 on the overarching principles of the Procurement Act B.E. 2560) are answered by introductory passages that appear to have been excluded from the indexed corpus or fragmented across chunk boundaries in a manner that obscures their relevance.
- Near-duplicate retrieval: A substantial number of retrieved sets contain duplicate or near-identical chunks (detected in 311 of 323 queries), consuming rank slots that could otherwise be used to surface the correct passage.

6.6 Discussion of Results

The retrieval evaluation results are broadly positive and confirm that the paraphrase-multilingual-MiniLM-L12-v2 model is well-suited to the procurement domain. The model's multilingual pre-training enables it to handle Thai-language regulatory text without domain-specific fine-tuning, and the overall Recall@6 of 87.31 percent is sufficient to support high-quality downstream generation for the majority of user queries.

However, three areas warrant further engineering attention before the system is considered production-ready at scale:

- Deduplication: The high prevalence of duplicate cosine scores (96.3 percent of queries) indicates that the same document chunk is being indexed multiple times or that chunk boundaries produce highly similar overlapping segments. Implementing a maximum marginal relevance (MMR) or hash-based deduplication step at retrieval time would improve precision and ensure greater diversity in the retrieved context window.
- Chunk boundary optimisation: The 12.7 percent complete-miss rate is concentrated in queries that target introductory, definitional, or cross-sectional content. Revisiting the document segmentation strategy — for example, using hierarchical or sentence-level

chunking for preamble sections — may recover a meaningful proportion of these failure cases.

- Domain-adaptive fine-tuning: For specialised procedural queries involving electronic form field codes and numerical procurement thresholds, lightweight fine-tuning of the embedding model on domain-specific question-passage pairs could improve cosine alignment between query and document without requiring a full model replacement.